

Complex Analysis – 2

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Liouville's theorem and its polynomial version; the little Picard theorem; the Schwarz lemma and Schwarz–Pick; automorphisms of the disc; the maximum modulus principle and Hadamard's three-lines lemma; the Blaschke condition $\sum(1-|a_n|) < \infty$ and finite Blaschke products; the identity theorem and the reflection principle; the argument principle as a way of counting zeros.

1. (Высшая лига 2020) Let f be an entire function which maps every unbounded sequence to an unbounded sequence. Is it true that f is a polynomial?
2. (Высшая лига 2022) Find all entire functions on $\mathbb{C}$ such that $f(T_r) = T_r$ for every $r > 0$, where $T_r = \{z : |z| = r\}$.
3. (Высшая лига 2024) Let $f: \mathbb{D} \rightarrow \mathbb{C}$ be holomorphic on the unit disc and suppose $f(z) \in \mathbb{R}$ whenever $\arg z = \pm \frac{\pi}{3}$. Prove that $f(\frac{1}{2}) \in \mathbb{R}$.
4. (S.T. Yau CSMC 2015) Let $\Delta = \{z : |z| < 1\}$ and let $f: \Delta \rightarrow \Delta$ be analytic and extendable continuously to an arc $I \subset \partial\Delta$. Prove that if $f|_I$ is constant, then f is constant on Δ . Does the analogous statement hold when Δ is replaced by a Jordan domain?
5. (S.T. Yau CSMC 2021) Let P be a polynomial with complex coefficients. Prove that the equation $e^z = P(z)$ has infinitely many solutions in $\mathbb{C}$.
6. (S.T. Yau CSMC 2021) Let f be a bounded holomorphic function on the strip $B = \{z : 0 < \operatorname{Re} z < 1\}$ which extends continuously to $\overline{B}$, and put $A_0 = \sup_{\operatorname{Re} z=0} |f(z)| > 0$, $A_1 = \sup_{\operatorname{Re} z=1} |f(z)| > 0$. Prove that $|f(z)| \leq A_0^{1-\operatorname{Re} z} A_1^{\operatorname{Re} z}$ for all $z \in B$.

7. (S.T. Yau CSMC 2022) Let $U \subset \mathbb{C}$ be a non-empty open set and let $f: U \rightarrow U$ be a non-constant holomorphic function with $f \circ f = f$. Prove that $f(z) = z$ for all $z \in U$.
8. (VJIMC 1997) Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be holomorphic with $|f(z)| = 1$ for all $|z| = 1$. Prove that there are $\theta \in \mathbb{R}$ and $k \in \{0, 1, 2, \dots\}$ with $f(z) = e^{i\theta} z^k$ for all z .
9. (VJIMC 2019) Let $D = \{z \in \mathbb{C} : \operatorname{Im} z > 0, \operatorname{Re} z > 0\}$, let $n \geq 1$ and let $a_1, \dots, a_n \in D$ be distinct. Define $f(z) = z \prod_{j=1}^n \frac{z-a_j}{z-\bar{a}_j}$. Prove that f' has at least one root in D .
10. (VJIMC 2025) Let $\mathbb{D} = \{z : |z| < 1\}$ and let $f: \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $\lim_{|z| \rightarrow 1} |f(z)| = 1$. If $f(z) = \sum_{n \geq 0} a_n z^n$, prove that $\sum_{n \geq 0} n|a_n|^2$ equals the number of zeros of f counted with multiplicity.
11. (IMC 2009) Let $p(z) = a_0 + a_1 z + \dots + a_n z^n$ be a complex polynomial and let $1 = c_0 \geq c_1 \geq \dots \geq c_n \geq 0$ be a convex sequence, that is $2c_k \leq c_{k-1} + c_{k+1}$ for $k = 1, \dots, n-1$. Put $q(z) = c_0 a_0 + c_1 a_1 z + \dots + c_n a_n z^n$. Prove that $\max_{|z| \leq 1} |q(z)| \leq \max_{|z| \leq 1} |p(z)|$.